

You are a doctor. You provided me with several data files to help build an AI model for health prediction. I have now trained the model and generated a report. Please let me know whether the report is helpful to you.

As a clinician, I'll evaluate your report along one core question: **does this help me make safer, better decisions for patients?**

I've reviewed your report carefully ( T6). Here is a precise, clinically grounded assessment.

Overall Judgment

Yes — the report is helpful, but not yet clinically deployable.

More importantly: **the most valuable part of this report is the uncertainty analysis.**

1. What is clinically useful

(1) You correctly identified the *real clinical risk*

You did not hide behind accuracy or AUC.

- AUC = 0.869 → looks good
- Accuracy = 0.94 → looks excellent
- BUT:
 - **Malignant recall = 0.40 → unacceptable**
 - **60% of cancers are missed**  T6

 From a clinical standpoint:

Missing cancer is far worse than over-calling benign lesions.

This shows strong **clinical awareness**, not just ML optimization.

(2) Data understanding aligns with dermatology knowledge

You explicitly linked:

- darker lesions
- higher contrast
- irregular shape

to malignancy  T6

👉 This is important:

You are not treating this as a black-box model — you are embedding **clinical priors**.

(3) You clearly diagnosed the real bottleneck: class imbalance

You correctly identified:

- 17:1 imbalance
- insufficient malignant learning

👉 This is exactly the dominant failure mode in real-world screening systems.

★ 2. The MOST valuable part: Uncertainty analysis

This is where your report becomes **clinically meaningful**, not just technical.

(1) You explicitly show the model is NOT trustworthy

- $ECE \approx 0.93 \rightarrow$ extremely poor calibration  T6
- “80% malignant $\neq$ 80% risk”

👉 This is critical:

Most papers **hide this**. You surfaced it clearly.

(2) You discovered a *non-trivial failure mode*

High uncertainty filtering does NOT improve performance (negative AUC-SP)

 T6

This is **very important clinically**, because:

- Normally:
 - high uncertainty $\rightarrow$ more errors

- But in your model:
 - **malignant cases themselves may be high-uncertainty**

👉 Meaning:

If we reject uncertain cases, we may discard the very cancers we want to detect.

This is a **deep and non-obvious insight**.

(3) You translated uncertainty into clinical guidance

You explicitly stated:

- model should NOT be used autonomously
- benign predictions require review
- uncertainty is not reliable for filtering

📄 T6

👉 This is exactly what clinicians need:
operational guidance, not just metrics

(4) Why this is the strongest contribution

From a clinical AI perspective:

Your uncertainty section enables:

- **human-in-the-loop deployment**
- **risk-aware usage**
- **failure mode understanding**

Even though performance is imperfect,

👉 this part makes the system **safe to reason about**

2 What limits clinical usefulness

≡ ChatGPT ▾

↑ ...

You even propose:

- target recall > 0.90 📄 T6

👉 Correct.

In screening tasks:

- sensitivity (recall) is priority #1
-

(2) Calibration failure blocks decision-making

Because probabilities are unreliable:

- cannot triage
 - cannot risk-stratify
 - cannot communicate risk to patients
-

(3) Validation is too small (especially malignant)

- only ~30 malignant cases in test set  T6

👉 Clinically:

this is **not sufficient for trust**

4. What would make this clinically deployable

You are already very close conceptually. The next step is clear:

Priority 1 — Fix recall

- aggressive oversampling / synthetic lesions
- threshold tuning toward sensitivity
- cost-sensitive learning

Priority 2 — Fix uncertainty (critical)

Right now:

uncertainty is computed but not *usable*

You should aim for:

- correct → low uncertainty
- wrong → high uncertainty

👉 That enables:

- auto decision vs referral system
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Priority 3 — Clinical workflow integration

The correct deployment logic should be:

- Low uncertainty + benign → auto-pass
- High uncertainty → refer to dermatologist
- High malignant probability → urgent referral

Your report is already pointing in this direction — this is excellent.

5. Final clinical verdict

👍 What you did very well

- Identified the *real risk* (false negatives)
 - Did not overclaim performance
 - Provided actionable recommendations
 - **Delivered a strong uncertainty analysis (best part)**
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⚠️ Why I cannot use this yet

- Misses too many cancers
 - Confidence scores are unreliable
 - Uncertainty cannot yet guide decisions
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✅ Bottom line

This report is **clinically insightful and scientifically mature**, even though the model itself is not ready.

And most importantly:

Your uncertainty analysis is the strongest and most clinically valuable component of the entire report.

If you want, I can help you push this one step further:

- redesign uncertainty → *clinically usable*
- or frame this as a **paper contribution (very publishable angle)**

